

Week 7 Unification Algorithm

Solve the following

- 1) Find Most general Unifier (MGU) of $\{P(b, x, f(g(z))) \text{ and } P(z, f(y), f(y))\}$

$$\Rightarrow \begin{aligned} & \bullet b = z \rightarrow z = b \\ & \bullet x = f(y) \\ & \bullet f(g(z)) = f(y) \rightarrow g(z) = y \end{aligned}$$

Substitution

$$\begin{aligned} & \bullet z \rightarrow b \\ & \bullet x \rightarrow f(y) \\ & \bullet y \rightarrow g(z) \end{aligned}$$

$$\text{MGU: } \{z/b, x/f(y), y/g(z)\}$$

- 2) Find most general Unifier (MGU) of $\{Q(a, g(x, a), f(y)) \text{ and } Q(a, g(f(b), a), x)\}$

$$\Rightarrow \begin{aligned} & \bullet a = a \\ & \bullet g(x, a) = g(f(b), a) \rightarrow x = f(b) \\ & \bullet f(y) = x \rightarrow f(y) = f(b) \\ & \bullet x \rightarrow f(b) \\ & \bullet y \rightarrow b \end{aligned}$$

$$\text{MGU: } \{x/f(b), y/b\}$$

- 3) Find MGU of $\{P(f(a), g(y)), P(x, x)\}$

$$\Rightarrow \begin{aligned} & f(a) = x \\ & g(y) = x \end{aligned}$$

$f(a) = g(y) \rightarrow$ which cannot unify
 $\therefore$ Therefore, no unifier exists.

4) Unify $\langle \text{prime}(11) \ \& \ \text{prime}(y) \rangle$
 $\Rightarrow$ $\cdot 11$ is a constant, y is a variable
 $\cdot \text{prime}(11) = \text{prime}(y) \rightarrow y = 11$
 Memo: $\langle y/11 \rangle$

5) Unify $\langle \text{knows}(\text{John}, x), \text{knows}(y, \text{mother}(y)) \rangle$
 $\Rightarrow$ $\cdot \text{knows John} = y \rightarrow y = \text{John}$
 $\cdot x = \text{mother}(y) \rightarrow x = \text{mother}(\text{John})$
 Memo: $\langle y/\text{John}, x/\text{mother}(\text{John}) \rangle$

6) Unify $\langle \text{knows}(\text{John}, x), \text{knows}(y, \text{Bill}) \rangle$
 $\Rightarrow$ $\cdot \text{knows John} = y \rightarrow y = \text{John}$
 $\cdot x = \text{Bill}$
 Memo: $\langle y/\text{John}, x/\text{Bill} \rangle$

Unification Algorithm.

Step 1: If ψ_1 or ψ_2 is a variable or constant, then:

- If ψ_1 or ψ_2 are identical, then return NIL
- Else if ψ_1 is a variable,
 - then if ψ_1 occurs in ψ_2 then return FAILURE
 - Else return $\langle \psi_2/\psi_1 \rangle$

- Else if ψ_2 is a variable,
 - if ψ_2 occurs in ψ_1 then return FAILURE,
 - Else return $\langle \psi_1/\psi_2 \rangle$

- Else return FAILURE

Step 2: If the initial predicate symbol in ψ_1 & ψ_2 are not same, then return FAILURE

Step 3: If ψ_1 & ψ_2 have a different number of arguments, then return FAILURE

Step 4: Set substitution set (SUBSET) to NIL

Step 5: For $i = 1$ to the number of elements in ψ_1

- a) Call Only function with i th element of ψ_1 & i th element of ψ_2 , & put the result into S
 - b) If $S = \text{failure}$ then return S Failure
 - c) If $S \neq \text{NIL}$ then do,
 - a. Apply S to the remainder of both L_1 & L_2
 - b. $\text{SUBST} = \text{APPEND}(S, \text{SUBST})$
- Step 6: Return SUBST

o/p :
 Map : $\langle 'b' : 'z', 'x' : ('y', 'y'), 'y' : ('g', 'z') \rangle$

MGU: {'b': 'z', 'X': ('f', 'Y'), 'Y': ('g', 'Z')}

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